



Hydrogen-producing enzymes

A team has discovered how hydrogenases are activated at the time of their biosynthesis. <https://digit.in/aug19-03>



Mosquitoes wiped out!

On two Chinese Islands. Scientists have managed to eradicate mosquitoes here in just 2 years. <https://digit.in/aug19-04>

three spatial dimensions or four fundamental forces?

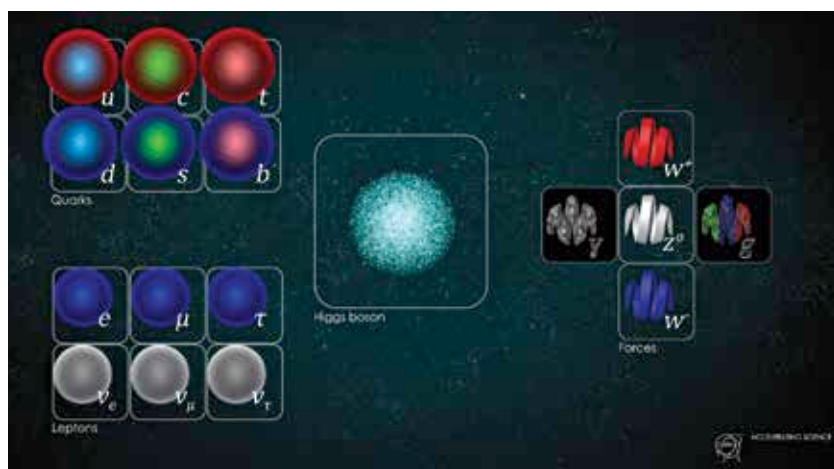
THE STANDARD MODEL

If you are familiar with the Standard Model and the various types of subatomic particles, you can just skip this subheading and move on to the next one. All matter is divided into the building blocks called fermions, or the force carrying particles known as bosons. The fermions are further divided into two basic groups called quarks and leptons. Both the quarks and leptons have three generations, with the particles becoming heavier in the second and third generations. The first generations are the most stable particles, with the heavier particles tending to decay into the first generation versions.

The quarks are up, down, charm, strange, top and bottom. Then there are the leptons. The electron, the muon and tau, as well as the electron neutrino, muon neutrino and the tau neutrino. While fermions cannot occupy the same space at the same time, the same restriction is not there for the bosons. Satyendra Bose and Albert Einstein together described the rules that bosons follow, known as Bose-Einstein statistics.

Now, there are the names for composite particles, that are group names for composite particles of a particular type. Mesons are made up of a quark and antiquark pair, and include the kaon and the pion. Baryons have at least three quarks (and we know of only the ones that have exactly three so far). The most common examples are protons and neutrons. Regular matter is referred to as baryonic matter. Baryons and mesons together are called hadrons, which are composite particles made up of quarks.

Now the quarks are held together within the baryons by particles known as gluons. The gluons and quarks have colour charges, which is just a way to describe three flavors of the particles. The most widely accepted theory that describe the rules of the ways in which quarks and gluons stick together is



The Standard Model

Image: CERN

known as quantum chromodynamics. However, there are other theories as well, such as the technicolor theory, which got ruled out because of the discovery of the Higgs Boson. However, the advancement of the technicolor theory continues, with consideration that the Higgs Boson could be a composite state.

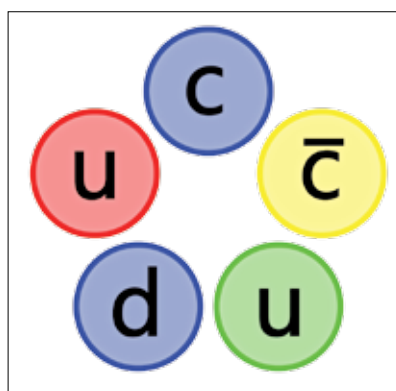
GLUEBALL

Gluons are the mediating particles of the strong force between quarks inside hadrons such as protons and neutrons. Like the quarks, gluons carry the colour charge for the strong interactions. Theoretically, the gluons could form clumps of their own because of the colour charge that they carry, and interact with each other, without any quarks being involved. The Super Proton Synchrotron and the Low Energy Antiproton Ring at CERN have both conducted experiments to

try and identify glueballs, by using streams of pions, which are particles made up of quarks and antiquarks. In the resulting collision events, two or three gluons can form bound states independent of the quarks, forming the glueballs. However, any potential glueballs would be difficult to differentiate from other mesons, which are also made up of pairs of quarks and antiquarks. The GlueX experiment at the Thomas Jefferson National Accelerator Facility is actively looking for any exotic mesons, including glueballs. Such matter is known as quarkless matter. According to the current understanding of physics, nearly all the properties of glueballs can be determined based on what we know of quarks and antiquarks. Additionally, researchers can also determine how the gluons are likely to clump up, and identify potential glueball states. Over the years, there has been evidence of a number of glueball candidates, but none have been positively identified as such, so far.

TETRAQUARKS

When it comes to hadrons, baryons such as protons and neutrons have three quarks within them, while mesons are made up of a quark-antiquark pair. The underlying theories of physics allow for exotic composite subatomic particles with four quarks. Candidate tetraquarks have been observed in experiments at CERN, Fermilab



A pentaquark candidate from 2015